

Omar Alkhadra

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EDUCATION

University of Southern California

Master of Science in Applied Data Science (4.0/4.0)

Los Angeles, CA

Aug 2023 – May 2025

George Washington University

Bachelor of Science in Systems Engineering, Minor in Statistics (3.64/4.0)

Washington, D.C.

Aug 2019 – May 2023

EXPERIENCE

GEICO - Pricing

Data Scientist I

Chicago, IL

Jun 2025 – Present

- Developing XGBoost model on 150M+ policyholder records, predicting defection probabilities as a function of premium changes and customer features; hyperparameter tuned with regularization for production deployment
- Built stochastic simulation of policyholder attrition under rate change scenarios with interactive dashboard; adopted for pricing decisions across states representing >\$100M in annual written premium

University of Southern California

Machine Learning Researcher

Los Angeles, CA

Dec 2024 – May 2025

- Co-authored paper published at ACM SIGSPATIAL '25 on probabilistic assignment of noisy GPS trajectories to candidate locations under spatial uncertainty
- Reconstructed proprietary probabilistic baselines (SafeGraph's multi-stage KDE pipeline, Nishida's ILP formulation with spatiotemporal priors) from limited documentation to establish reproducible benchmarks

George Washington University

Research Assistant

Washington, D.C.

Aug 2022 – May 2023

- Co-authored peer-reviewed paper in *Environment Systems and Decisions*; built dynamic inoperability input-output model (Leontief) to quantify economic losses across interdependent sectors facing drought
- Conducted sensitivity analysis of policy intervention scenarios with expert-elicited parameters; validated model outputs against published economic loss estimates

PROJECTS

Probabilistic Forecasting: Prediction Market Strategy

- Developed calibrated probability ML model using forecast error prediction with Student-t residual distributions, multi-source NWP weather and atmospheric features to price temperature contracts against live market odds
- Backtested over 14-month out-of-sample period across 10 cities (1,012 trades): 49.8% ROI/trade, 4.9 Sharpe, 5.6% max drawdown; Monte Carlo simulation (25K paths) confirms 99.1% path profitability, 3.2 median Sharpe
- Validated calibration via PIT testing (KS=0.033) & Diebold-Mariano test confirms model outperforms market (p=0.006) on selected trades. Deployed live with automated daily execution: 2.86 Sharpe over 314 trades

Volatility Forecasting for Delta-Neutral Options Trading

- Engineered volatility forecasting framework comparing XGBoost, GARCH(1,1), and historical baselines; Diebold-Mariano test confirms XGBoost outperforms GARCH (p=0.004) with 21% lower RMSE
- Implemented full delta-neutral options backtesting engine across 5 years of SPY options data with Black-Scholes pricing, daily delta rebalancing, and regime-conditional VRP estimation

SKILLS

- Programming:** Python (pandas, NumPy, SciPy, XGBoost, scikit-learn), SQL, R
- Data & Tools:** Git, Linux, BigQuery, Jupyter, REST APIs, AWS, Docker, Dash, Plotly
- Quantitative:** Feature Engineering, Statistical Modeling, Data Analysis, Machine Learning, Monte Carlo Methods

PUBLICATIONS

- Saxena, N., Hsu, S., Shetty, M., **Alkhadra, O.**, Shahabi, C., & Horn, A. L. (2025). POIFormer: A Transformer-Based Framework for Accurate and Scalable Point-of-Interest Attribution. *SIGSPATIAL '25*, 607–619. ACM.
- Santos, J. R., **Alkhadra, O.**, Makrides, S., Huang, Y., & Menta, A. (2025). Drought impacts across economic sectors with policy analysis of mitigations. *Environment Systems and Decisions*.